POSTNIKOV TRIPLES, THEIR MAPS, AND THE SPHERES

Abstract. Pointed simply connected 3-types are classified by Postnikov triples (A, B, k) with $k : K(A, 2) \to_* K(B, 4)$. We show that arbitrary pointed maps between them are classified as well: pointed maps $\mathrm{Fib}k \to \mathrm{Fib}k'$ are equivalent to morphisms of triples, pairs of homomorphisms with a homotopy. Triples and the homotopy classes of their morphisms then form a precategory isomorphic to the homotopy precategory of pointed simply connected 3-types. On this algebraic side the universal properties of S^2 and S^3 become computations with the cohomology of Eilenberg–MacLane spaces, giving $\pi_3(S^2) \cong \mathbb{Z}$ and $\pi_4(S^3) \cong \mathbb{Z}/2$. The last section says what is formalised.

Attribution. The strategy is David Wärn's: classify pointed simply connected 3-types *and their arbitrary pointed maps* by Postnikov data, identify the universal objects of the spheres on the side of the data, and treat $\pi_4(S^3)$ by embedding pointed 2-connected 4-types into pointed simply connected 3-types by looping. The classification of objects is Hart–Milner's (Owen Milner's equivalence). We do not know Owen Milner's unpublished treatment of maps; nothing here claims to reproduce it.

1. The plan

To compute $\pi_3(S^2)$ it suffices to know a pointed simply connected 3-type P with a class $s \in [S^2, P]_*$ such that

$$s^* : [P, X]_* \to [S^2, X]_*$$

is a bijection for every pointed simply connected 3-type X: then P is the 3-truncation of S^2, and $\pi_3(S^2) = \pi_3(P)$ by a Yoneda argument. Similarly, $\pi_4(S^3)$ is computed from a pointed 2-connected 4-type with the analogous universal property among such types.

Wärn's idea is to make the left-hand side of this bijection *algebraic*. If pointed simply connected 3-types and their maps are described by algebraic data, then $[P, X]_*$ becomes a set of classes of algebraic morphisms, computable from the cohomology of Eilenberg–MacLane spaces, while $[S^2, X]_*$ is computed from the sphere. The argument has three steps:

(1) classify the objects (Hart–Milner);
(2) classify the arbitrary pointed maps (Theorem 1);
(3) compute the sphere's universal property on the algebraic side (Sections 5 and 6), and recover the homotopy group.

2. Objects

A *Postnikov triple* is $t = (A, B, k)$ with abelian groups A, B and a pointed map $k : K(A, 2) \to_* K(B, 4)$. Its realisation is the fibre

$$\mathrm{Fib}k = \textstyle\sum_{x:K(A,2)} (k\, x = \mathrm{pt}),$$

a pointed simply connected 3-type with $\pi_2 = A$ and $\pi_3 = B$. The map

$$\mathsf{realise} : \mathsf{Triples} \to \mathsf{SC3}$$

is an equivalence of types. Both types live in the next universe and are not sets; they are 2-types.

unit/doc/postnikov-morphisms-overview.tex -- compiled November 5, 2026, 14:21

3. Maps

For triples $t = (A, B, k)$ and $t' = (A', B', k')$ define

$$\mathsf{Mor}(t, t') = \sum_{a:\mathrm{Hom}(A,A')} \; \sum_{b:\mathrm{Hom}(B,B')} \big(K(a,2)\,;\, k' = k\,;\, K(b,4)\big).$$

Here $f\,;\,g$ means "f, then g", and the last component is a *path of pointed maps*: a homotopy together with its coherence at the base point. Such a morphism determines a map of fibres, $(x, q) \mapsto (K(a,2)\,x,\ H_x \cdot K(b,4)(q))$, the *canonical map*.

Theorem 1. *For all triples t, t', the canonical map*

$$\mathsf{Mor}(t, t') \longrightarrow \mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$$

is an equivalence of types. There is no hypothesis on the groups or on k, k', and no sphere axiom.

Both sides are 1-types, not sets in general. The homotopy H is genuine data. For $t = (A, 0, 0)$ and $t' = (0, C, 0)$ the coefficients carry no information at all, and both sides are $\mathrm{Map}_*(K(A,2), K(C,3))$, whose components form $H^3(K(A,2); C)$.

Idea of the proof. A pointed map $\mathrm{Fib}k \to \mathrm{Fib}k'$ is a map to $K(A', 2)$ together with a pointed null-homotopy of its composite with k'.

- *The base.* The projection $\mathrm{Fib}k \to K(A,2)$ has 2-connected fibres, and $K(A', 2)$ is a 2-type. So maps into $K(A', 2)$ factor uniquely through $K(A, 2)$, and
$$\mathrm{Map}_*(K(A,2), K(A',2)) \simeq \mathrm{Hom}(A, A').$$
- *The key lemma.* For $c : K(A,2) \to_* K(B', 4)$, pointed null-homotopies of $\mathrm{Fib}k \to K(A,2) \xrightarrow{c} K(B',4)$ are equivalent to factorisations of c through k, that is, to pairs $(\beta,\ c = k\,;\,\beta)$.

In fibrewise terms, a null-homotopy is a family of maps $F_x : (k\,x = \mathrm{pt}) \to (c\,x = \mathrm{pt})$ between "$K(B,3)$-torsors". The proof separates a linear part from a translation:

- whether F_x is a translate of the map induced by a fixed β is a *proposition*, by the connectivity of the fibres and the truncation of the target;
- so it suffices to check it over the base point, since the base is connected;
- over the base point, the restriction of F is a pointed map $\Omega K(B,4) \to \Omega K(B',4)$, and it has a *unique* delooping β (Wärn's delooping theorem);
- the translations are exactly the homotopy component, which stays as data.

No point is chosen in any fibre other than the base, no representative of a class is chosen, and lifts are not assumed unique. Finally, $\mathrm{Map}_*(K(B,4), K(B',4)) \simeq \mathrm{Hom}(B, B')$. The composite of these equivalences is, by definition, the canonical map.

4. The homotopy precategory of triples

Composition of morphisms of triples is defined intrinsically. The coefficients compose, and the homotopies paste:

$$K(a\,;\,a',2)\,;\,k'' = K(a,2)\,;\,k'\,;\,K(b',4) = k\,;\,K(b\,;\,b',4).$$

The identity uses $K(1, m) = \mathrm{id}$. The canonical maps of fibres preserve composition and identities up to pointed homotopy, that is, in the homotopy classes $[\mathrm{Fib}k, \mathrm{Fib}k'']_*$. Put $\hom(t, t') = \|\mathsf{Mor}(t, t')\|_0$.

Theorem 2. *Triples, with the sets $\hom(t, t')$ and the induced composition and identities, form a precategory. The functor to the homotopy precategory of pointed simply connected 3-types has two parts:*

- *on objects it is* realise*, an equivalence of types;*
- *on hom-sets it is the class of the canonical map, a bijection by Theorem 1.*

It preserves composition and identities.

In the terminology of Ahrens, Kapulkin and Shulman (*Univalent categories and the Rezk completion*, Definition 6.9), this is an *isomorphism of precategories*: a fully faithful functor whose object map is an equivalence of types. A precategory may have an arbitrary type of objects (here 2-types), while its hom-types are sets. Neither precategory is univalent in their sense, and no such claim is made. By their Lemma 6.16, under univalence an isomorphism of precategories yields an equality of the two precategories. No explicitly declared inverse functor is needed for the criterion, and equality in $[\mathrm{Fib}k, \mathrm{Fib}k'']_*$ is all the homotopy precategory requires.

5. $\pi_3(S^2)$ on the triple side

Take $t_0 = (\mathbb{Z}, \mathbb{Z}, \kappa)$, with κ the cup square, and $P = \mathrm{Fib}\kappa$.

Proposition 3. *For every triple $t' = (A', B', k')$, the map $\hom(t_0, t') \to \mathrm{Hom}(\mathbb{Z}, A')$, $[a, b, H] \mapsto a$, is a bijection.*

The proof is a direct computation on Mor:

- (H4), the bilinear classification of $[K(\mathbb{Z}, 2), K(B', 4)]_*$, shows that for each a there is exactly one b for which $K(a, 2)\,;k'$ and $\kappa\,;K(b, 4)$ are homotopic;
- (H3), $[K(\mathbb{Z}, 2), K(B', 3)]_* = 0$, shows that the homotopies between them form a single component.

The second point needs loops at an arbitrary point of the space $\mathrm{Map}_*(K(\mathbb{Z}, 2), K(B', 4))$. Writing $K(B', 4) \simeq \Omega K(B', 5)$, pointwise multiplication by g^{-1} translates them to loops at 0.

On the sphere side, maps $S^2 \to \mathrm{Fib}k'$ correspond to maps $S^2 \to K(A', 2)$, hence to $\mathrm{Hom}(\mathbb{Z}, A')$. The marked class $s : [S^2, P]_*$ makes the triangle commute, because the canonical map of fibres lies over $K(a, 2)$. Therefore $s^* : [P, \mathrm{Fib}k']_* \to [S^2, \mathrm{Fib}k']_*$ is a bijection for every triple. By the classification of objects, this is the universal property for every pointed simply connected 3-type, and $\pi_3(S^2) \cong \mathbb{Z}$ follows.

6. $\pi_4(S^3)$ on the triple side

Looping sends a pointed 2-connected 4-type Y to the pointed simply connected 3-type ΩY; it is an embedding SC4 $\hookrightarrow$ SC3. Its image consists of the *deloopable* triples. On maps, looping is a bijection $[X, Y]_* \simeq [\Omega X, \Omega Y]_*$. The candidate is $\bar{t} = (\mathbb{Z}, \mathbb{Z}/2, \bar{k})$, with $\bar{k}$ the cup square reduced mod 2, and $R = \mathrm{Fib}\bar{k} = \Omega BR$.

Proposition 4. *For every deloopable triple t', the map $\hom(\bar{t}, t') \to \mathrm{Hom}(\mathbb{Z}, A')$ is a bijection.*

Again the computation is on Mor:

- (H4) and Statement T (deloopability forces $2h(1) = 0$) show that the required class factors through $\mathbb{Z}/2$;
- injectivity of $g \mapsto [\bar{k}\,;K(g, 4)]$ fixes b;
- (H3) controls the homotopies, as in Section 5.

On the sphere side, the source is ΩS^3. Lifting there needs an existence hypothesis, and it is obtained from the surjectivity in the proposition above. The looped marked class then has the universal property against every ΩY, and by the looping embedding statement M holds for every pointed 2-connected 4-type. Both instances used in the final step come from it: BR itself and $\|S^3\|_4$. This gives $\pi_4(S^3) \cong \pi_4(BR) \cong \mathbb{Z}/2$.

7. What is formalised

Everything above is checked in Unit, in `sag/pi3-s2-postnikov`, with univalence and replacement as parameters and the universal property of the sphere as the parameter telescope of a module; the detailed informal proof, with a declaration-by-declaration correspondence, is `doc/postnikov-morphisms.pdf`.

- Objects: `PostnikovClassification.classification`.
- Theorem 1: `TripleMapsEquiv.Maps2.realise-map-equiv`; the canonical map is `FibMap.fm`, the key lemma `TripleMaps.Key.key`, the delooping theorem `DeloopMaps`.
- Theorem 2: `TripleHomotopy.Cat`, with composition and identities `TripleHomotopy.Maps3.rmap-comp`, `Maps1.rmap-id`.
- $\pi_3(S^2) \cong \mathbb{Z}$: `Pi3TripleUniversal.Final.pi3-S2`, under the S^2 axiom only, with `TripleComponents` for the translation of loops.
- $\pi_4(S^3) \cong \mathbb{Z}/2$: `Pi4TripleUniversal.Final4.pi4-S3`, under the S^3 axiom only.

These proofs reuse the cup square, (H3), (H4), Statement T, the delooping embedding and the computations on the sphere side, but not the earlier, specialised mapping theorems of the development, which remain as references. The development checks in a fresh process: 5633 declarations, 0 failed.

Two points are not formalised: that the coefficient b of the canonical map is the map induced on π_3 through $\pi_3(\mathrm{Fib}k) \cong B$ (the formal statement says that the canonical map lies over $K(a,2)$ and restricts to $\Omega K(b,4)$ on the base fibre), and the consequence, by Ahrens–Kapulkin–Shulman's Lemma 6.16, that the two precategories are equal.

The unfolding specifications (`.opens`) of the six new modules were generated from the checker and audited against the informal proof by kind; the entry-by-entry audit remains to be done.